

Harish Anculgari

Full Stack Developer and Data Science Graduate

Phone: +91 9177454533 Email: harishpandu548@gmail.com **Portfolio:** harish-h-portfolio.vercel.app

LinkedIn: linkedin.com/in/harish-pandu-018427280

GitHub: github.com/harishpandu548

Summary

Motivated and detail focused Full Stack Developer and Data Science Graduate with experience in building modern web applications. Strong exposure to artificial intelligence driven systems, secure authentication workflows, and production ready deployments using current web technologies.

Technical Skills

Frontend Technologies: HTML, CSS, JavaScript, TypeScript, React, Nextjs, Redux Toolkit (RTK), Context API, Tailwind CSS, Responsive UI/UX Design

Backend Technologies: Nodejs, Express.js, REST APIs, Multi-Tenant Architecture, Authentication & Authorization, Socket.IO, session management, JWT, NextAuth, OAuth (Google, GitHub, Facebook)

Databases & ORMs: MongoDB, PostgreSQL, Prisma ORM, Mongoose, Redis

Cloud, Deployment & DevOps: AWS EC2, Linux (Ubuntu), Nginx, PM2, GitHub Actions, CI/CD, VPS Management, SSL Configuration, Reverse Proxy Setup, Vercel, Render, Docker, Railway, Netlify

AI & Automation: LangChain, RAG (Retrieval-Augmented Generation), Prompt Engineering, Gemini AI API Integration, Multi-Agent AI Systems

Payments Integrations: Razorpay payment integration, webhook handling, Cloudinary, Multer

Developer Tools: Git, GitHub, Postman, Thunder Client, Axios, environment configuration

Programming Languages: Python, Java, data structures and algorithms

Experience

Tyro Digital - Full Stack Developer

Jan 2026 – Present

- Developed scalable full-stack applications with Next.js, TypeScript, Prisma, and PostgreSQL following modern production-grade architecture practices.
- Contributed to building systems involving user management, scheduling workflows, and administrative dashboards in domain-specific applications.
- Implemented secure authentication and role-based access control using JWT, NextAuth.
- Integrated AI-powered workflows and automation features using LLM APIs to improve user interactions and data processing efficiency.
- Optimized backend performance through efficient query design, indexing strategies. Participated in deployment and production infrastructure workflows involving Linux servers, CI/CD pipelines, and cloud-based hosting environments.
- Built payment, subscription, and credit-based workflows through third-party integrations such as Razorpay.
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Alrise – Full Stack Developer

Dec 2024 - Dec 2025

- Developed and maintained full-stack web applications using React, Node.js, Express.js, and database-driven architectures.
- Built and integrated RESTful APIs, authentication systems, and secure user access workflows for scalable applications.
- Worked on frontend optimization, reusable component design, backend business logic, and efficient data handling.
- Contributed to deployment, production environment setup, debugging, and application performance optimization using platforms like Netlify, Vercel, Render, Railways.

Projects

Hospital Management System – (Next.js, TypeScript, Prisma, PostgreSQL, Redis, JWT, Zod, Docker)

Project Link: github.com/harishpandu548/HospitalManagementSystem

- Architected a modular monolith backend for a hospital management system with strict role-based access control (RBAC), supporting ADMIN, RECEPTIONIST, DOCTOR, and PATIENT roles with active role context switching.
- Designed a permission-driven authorization system (user → role → rolePermission → permission) ensuring

fine-grained access control across all APIs, eliminating role hardcoding and enabling scalable policy management.

- Built a rule-based appointment scheduling engine with Redis-backed slot locking and background job processing, preventing double booking and ensuring consistency under concurrent requests.
- Implemented transactional workflows using Prisma (\$transaction) for critical operations (appointment lifecycle, doctor onboarding, role assignment) to maintain strong data integrity and avoid partial writes.
- Developed audit logging and state machine-driven appointment status management, ensuring traceability and enforcing valid state transitions across the system.
- Integrated asynchronous job processing using Redis queues and worker processes for event handling (appointment creation, reminders, cleanup), decoupling real-time operations from background processing.

Artificial Intelligence Software as a Service Platform – (Nextjs, TypeScript, Prisma, PostgreSQL, NextAuth, Razorpay, Vercel)

Project Link: ai-saas-application-gamma.vercel.app

- Designed and developed an artificial intelligence software as a service platform offering chat assistance, content generation, and document summarization tools and many more(10+ AI tools).
- Implemented secure user authentication, controlled user access, and a usage credit system.
- Built multiple tools including an AI chat assistant, blog generator, email writer, and PDF summarizer.

GenAI Studio – Multi-Agent AI Platform – (Python, Streamlit, LangChain, Mistral AI, ChromaDB, HuggingFace)

Project Link: genai-studio.streamlit.app/

- Designed and deployed a production-grade multi-agent AI platform featuring three intelligent tools — a Deep Research engine, a PDF RAG Chat assistant, and a real-time City Intelligence agent.
- Engineered a 4-agent autonomous research pipeline (Search → Read → Write → Critique) using LangChain, capable of surfacing sources, scraping content, generating structured reports, and self-critiquing quality with a score.
- Built a Retrieval-Augmented Generation (RAG) system using ChromaDB vector store and HuggingFace sentence embeddings, enabling semantic search and grounded Q&A over any uploaded PDF.
- Integrated real-time external APIs (OpenWeatherMap, Tavily) into an AI agent that fetches live weather and breaking news for any city on demand.

Production VPS Deployment

– (AWS EC2, Ubuntu, Nginx, PM2, GitHub Actions, SSL, DuckDNS)

Project Link: aws-vpss.duckdns.org/

- Provisioned and configured a Linux-based AWS EC2 VPS server for hosting full-stack production applications.
- Configured Nginx as a reverse proxy for routing traffic to Node.js applications running on custom ports. Managed application processes using PM2 with automatic restart and background process handling.
- Implemented CI/CD workflows using GitHub Actions to automate deployment after code pushes to GitHub.
- Configured HTTPS using SSL certificates and connected a custom DuckDNS domain to the VPS server.
- Worked with Linux server management, firewall rules, SSH access, environment variables, and deployment debugging.

YouTube Clone Application – (React, Nodejs, Express, MongoDB, Mongoose, Cloudinary, Tailwind CSS)

Project Link: clonetube-frontend.vercel.app

- Built a video sharing application with features including uploads, likes, comments, subscriptions, and watch history. Managed user interactions such as likes and comments using backend APIs.
- Implemented API based video listing and dynamic video playback.

Certifications

AI for Beginners - HP LIFE | Data Science & Analytics - HP LIFE | Introduction to Cybersecurity Awareness | Wadhvani Foundation Certificate | Internship common aptitude test – LabMentix

Education

Bachelor of Technology in Data Science
Malla Reddy College of Engineering and Technology, Hyderabad

2022 to 2025